

"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

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BE IV (ME)

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FINAL B.E. EXAMINATION - 2024

MECHANICAL ENGINEERING

ES

(VII SEMESTER CBCS)

ME - 404 A : MANUFACTURING TECHNOLOGY

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

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- ✓ (a) Considering plane stress system, find magnitudes and directions of principal stresses, maximum shearing stress and corresponding normal stress. Also construct Mohr's circle for $\sigma_x = 750 \text{ N/mm}^2$; $\sigma_y = 150 \text{ N/mm}^2$; $\sigma_z = 0$ and $\tau_{xy} = 150 \text{ N/mm}^2$ 10

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(b) For problem given in question 1(a), find yield stress of the material according to:

(a) Tresca criteria

(b) Von-Mises Criteria 6

2/ (a) What is rolling in the context of the bulk deformation processes? Identify some of the ways in which force in flat rolling can be reduced. What is draft in a rolling operation? 10

(b) A series of cold rolling operations are to be used to reduce the thickness of a plate from 50 mm down to 25 mm in a reversing two-high mill. Roll diameter = 700 mm and coefficient of friction between rolls and work = 0.15. The specification is that the draft is to be equal on each pass. Determine (a) minimum number of passes required, and (b) draft for each pass? 6

3/ (a) What is forging? One way to classify forging operations is by the degree to which the work is constrained in the die. By this classification, name

the three basic types. What are the basic types of forging equipment? *namely open press* 10

(b) A hot upset forging operation is performed in an open die. The initial size of the workpart is: $D_0 = 25 \text{ mm}$, and $h_0 = 50 \text{ mm}$. The part is upset to a diameter = 50 mm. The work metal at this elevated temperature yields at 85 MPa ($n = 0$). Coefficient of friction at the die-work interface = 0.40. Determine (a) final height of the part, and (b) maximum force in the operation. 6

4/ (a) Highlight the importance of extrusion process and derive expression for frictional power loss and maximum force required in extrusion process. 10

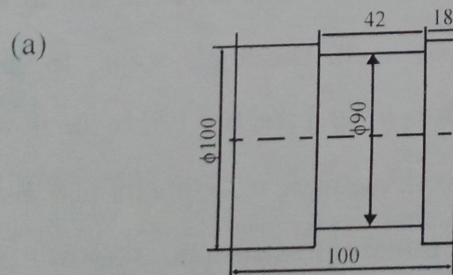
(b) A 3 inch long cylindrical billet whose diameter = 1.5 inch is reduced by indirect extrusion to a diameter = 0.375 inch. Die angle = 90° . In the Johnson equation, $a = 0.8$ and $b = 1.5$. In the flow curve for the work metal, $K = 75,000 \text{ lb/inch}^2$ and $n = 0.25$. Determine (a) extrusion ratio, (b) true

(Contd.)

strain (homogeneous deformation), (c) extrusion strain, (d) ram pressure, (e) ram force, and (f) power if the ram speed = 20 in/min. 6

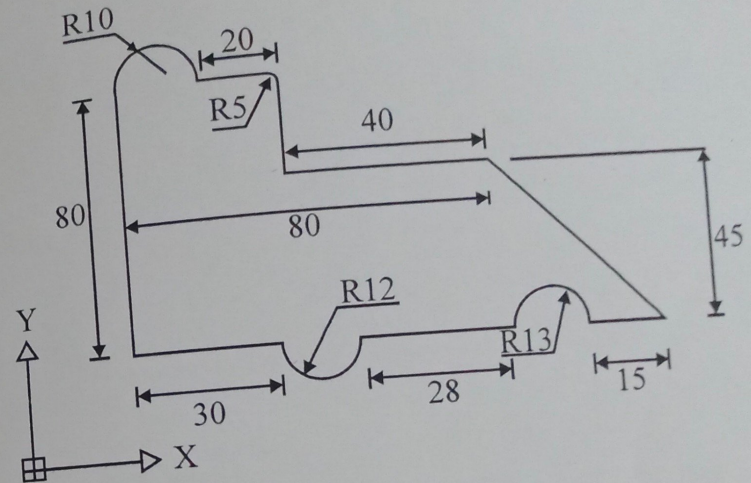
5. (a) Compare Electrochemical Machining and Electric Discharge Machining in terms of operating principle, key elements and process parameters. 8
- (b) Compare Electron Beam Machining and Laser Beam Machining in terms of operating principle, key elements and process parameters. 8
6. (a) Give details of three basic sheet metal processes: Cutting, Bending and Drawing with suitable examples. 8
- (b) Explain Flexible Manufacturing Systems and their applications. 8

7. Write CNC part program for following :



6

(b)



10

8. Write short notes on : (any four)
- (a) Robotics and Mechatronics
- (b) Manual and Computer Assisted CNC Part Programming
- (c) Computer Integrated Manufacturing Systems
- (d) Plastic Deformation and Yielding Criteria
- (e) Hot and Cold Working
- (f) Plasma Arc Machining
- (g) Ultrasonic Machining

4x4 = 16

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Roll No: 20VMECH1168

BE IV (M)

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FINAL B.E. EXAMINATION, 2023
MECHANICAL ENGINEERING
(VII SEMESTER CBCS)
ME 405 A : OPERATIONS RESEARCH

Time – Three Hours
Maximum Marks – 80

Note: (i) Attempt any **Five** questions.
(ii) Answer should be to the point.
(iii) All questions carry equal marks.

1. (a) Define the Meaning of operation Research? Write the
Modelling Process in OR? 6
(b) Solve the following Linear Programming Problem for
Maximization of Profit. 10

$$\text{Max } Z = 30x_1 + 20x_2$$

$$2x_1 + x_2 \leq 90$$

$$x_1 + 2x_2 \leq 80$$

$$x_1 + x_2 \leq 50$$

$$x_1, x_2 \geq 0$$

2. (a) Formulate the Problem.

The handy Dandy Company wishes to schedule the Production of a kitchen appliance that requires two resources - labour and Material. The Company is considering three different Models and its Production Engineering department has furnished the following data:

	Model		
	A	B	C
Labour (hours per unit)	7	3	6
Material (Pounds per unit)	4	4	5
Profit (\$ per unit)	4	2	3
	1		
	Cont.		

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The supply of raw material is restricted to 200 Pounds per day. The daily availability of Labour is 150 hours. Formulate a linear Programming Model to determine the daily Production rate of the various models in order to maximize the total Profit. 8

(b) Write the special Cases of LPP

(i) Infinite or Multi optimum solution

(ii) un bounded Solution

(iii) No solution or Infeasibility

(iv) Degeneracy 8

3. (a) What are the Differences between PERT and CPM? 8

(b) For the following set of activities draw the Network diagram.

(i) Determine the Critical Path

(ii) Expected Project Completion time

Activity	Precedence	Time		
		T_a	T_m	T_b
A	-	4	6	8
B	A	5	7	15
C	A	4	8	12
D	B	15	20	25
E	B	10	18	26
F	C	8	9	16
G	E	4	8	12
H	D, F	1	2	3
I	G, H	6	7	8

4. (a) Find the basic feasible Solution for the given Transportation Problem with the help of Vogel's approximation method. 8

		Destination				Supply
		A	B	C	D	
Source	1	3	100	7	4	300
	2	2	6	5	9	400
	3	8	30	3	2	500
Demand		250	350	400	200	

Handwritten notes: P_1 , P_2 , P_3 , $3 \times 250 = 750$, $3 \times 180 = 540$, $3 \times 250 = 750$

(b) Solve the following assignment Problem for minimization of cost. 8

12	30	21	15
18	33	9	31
44	25	24	21
23	30	28	14

5. (a) Write the types of Inventories? 4

(b) Annual demand is 8000 units ordering cost $C_o = 1800$ Per/order holding cost is $C_h = 10\%$ of unit purchase price of Inventory. Determine the best order Size when the items can be purchase as given below. 12

Lot Size	Unit Price
1-999	220
1000-1499	200
1500-1999	190
2000 & above	185

6. (a) What do you understand by Customer's attitude? Write the representation of Quening Model? 6

(b) The no. of Person's arriving at a Service Centre is 8 customer/hr and service provider takes 5min on an average to service a customer then find:

- Average No. of customer's in a Queue and System
- Waiting time of the customer in the Queue and System
- Average length of Non Empty Queue. 10

7. Write short note: (any four)

- Monte Carlo technique
- Game theory
- Duality
- Sensitivity Analysis
- Applications of Inventory
- Opportunity cost

4X4 16

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Roll No. 210MECV1523

BE IV (ME)

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OR

FINAL B.E. EXAMINATION - 2024

MECHANICAL ENGINEERING

(VII SEMESTER CBCS)

ME - 405 A : OPERATIONS RESEARCH

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) In game of Matching Coins, Player A wins Rs 8/- in Both Coins Shows heads Rs 1/- if both are tails. Player B wins Rs 3/- when coins do not match. Given the choice of being Player A or B, which would you choose and what would be your strategy?

8

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1

(Contd.)

Game Theory

(Contd.)

(b) Formulate the Problem

A company has two grades of Inspectors, 1 and 2, who are to be assigned for a quality control inspection. It is required that at least 1800 pieces be inspected per 8-hour day. Grade 1 inspectors can check pieces at the rate of 25 per hour, with an accuracy of 98%. Grade 2 inspectors check at the rate of 15 pieces per hour with an accuracy of 95%. The wage rate of a Grade 1 inspector is \$4.00 per hour, while that of a grade 2 inspector is \$2.00. The company has available for the inspection job eight grade 1 inspectors, and ten grade 2 inspectors. The company wants to determine the optimal assignment of inspectors, which will minimize the total cost of the inspection. 8

2. (a) Let us solve by the Simplex Method the following problem :-

$$\text{Maximize : } Z = 3x_1 + 2x_2$$

$$\text{Subject to : } -x_1 + 2x_2 \leq 4$$

$$3x_1 + 2x_2 \leq 14$$

(Contd.)

$$x_1 - x_2 \leq 3$$

$$x_1 \geq 0, x_2 \geq 0$$

14

8

- (b) Write the special cases of LPP

(i) Infinite or Multi optimum solution

(ii) Unbounded solution

(iii) Degeneracy

(iv) No solution or infeasibility

8

3. (a) Define Customer's attitude. Explain all the types of attitude. 6

- (b) In a Machine shop certain type of Machine break down at an average rate of 6/hr and estimated cost of ideal Machine is Rs 20/hr to repairment A and B with different skills are considered to be hired as repairment. A takes 5 min on average to repair A machine and his salary is Rs 10/hr where B takes 4 min on an average to repair B machine and salary is Rs 18/hr which repairment service should be used and why assume the workshift of 8 hours? 10

(Contd.)

4. (a) Write the difference between PERT and CPM. 4
 (b) Consider a Project consisting of 7 jobs (A.....G) with following Precedence relations and time estimates.

Job	Precedence	Optimistic time	Most likely time	Pessimistic time
A	-	2	5	8
B	A	6	9	12
C	A	5	14	17
D	B	5	8	11
E	C,D	3	6	9
F	-	3	12	21
G	E,F	1	4	7

- (a) Draw the Project Network diagram.
 (b) Determine the expected duration and variance of each Job.
 (c) What is expected length of Project and its variance

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5. (a) Find the basic feasible solution for the following Transportation Problem

Warehouse					
Factory	w ₁	w ₂	w ₃	w ₄	
F ₁	19	30	50	10	7
F ₂	40	30	40	60	9
F ₃	40	8	70	20	18
	5	8	7	14	34

8

(Contd.)

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- 5 (b) Solve the Assignment problem

Machine						
Job	M ₁	M ₂	M ₃	M ₄	M ₅	
J ₁	10	11	4	2	8	
J ₂	7	11	10	14	12	22
J ₃	5	6	9	12	14	
J ₄	13	15	11	10	7	8

- (a) Write the types of Inventories. Define why we required Inventory in the system. 6
 (b) A manufacturing company requires 2000 units of a Material Per Year. Co is Rs 10 Per order C_r is 0.16 per unit/yr. Purchase price Rs 1/unit. Find the EOQ and TIC if a discount of 5% is available for order of 1000 unit should manufacturer accept this order? Also he purchases a single lot of 2000 units he has to pay rupees 0.93 per unit what purchase quantity you recommend? 10

7. Write the Short notes (Any four)

- (i) Game theory
 (ii) Duality
 (iii) Simulation
 (iv) Quening Model
 (v) Sensitivity Analysis

4x4 = 16

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Roll No. ...214MECL1153

BE IV (ME)

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FINAL B.E. EXAMINATION - 2024

MECHANICAL ENGINEERING

(VII SEMESTER CBCS)

ME-402 A: INTERNAL COMBUSTION ENGINES

Time - Three Hours

Maximum Marks - 80

- Note:- (i) Attempt any **Five** questions.
(ii) Answer should be to the point.
(iii) All questions carry equal marks.

1. (a) Describe the phenomenon of detonation or knocking in S.I. engine. On what factors does detonation depend? 10
(b) With a neat sketch explain working of Multi Point Fuel Injection System (MPFI). 6
2. (a) Explain the process of combustion in CI Engine compare it with the combustion of SI Engine. 10
(b) What are the three methods of generating swirl in CI engine combustion chambers? 6
3. A test on a single cylinder, four stroke diesel engine having bore 18cm and stroke 36 cm yielding the following results speed 285 rev/min; brake torque 393 Nm. indicated mean effective pressure 7.2 bar; fuel consumption 3.5 kg/hr;

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cooling water flow 4.5 kg/min; cooling water temp^r rise 36°C; air - fuel ratio by mass 25; exhaust gas temp^r; 415°C; barometric pr. 1.013 bar; room temp^r 21°C.

The fuel has a calorific value of 45200 kJ/kg. Determine.

- (a) The indicated thermal efficiency; 5
- (b) The volumetric efficiency based on atmospheric conditions. 5

Draw up a heat balance in terms of kJ/min explaining clearly the content of such term.

Take $R = 0.287 \text{ KJ/Kg K}$

$C = 1.005 \text{ KJ/kg K}$ for exhaust gases. 6

4. (a) Explain with suitable sketches the following scavenging system.

- (i) Uniflow Scavenging
- (ii) Cross flow Scavenging
- (iii) Loop or reverse scavenging 12
- (b) Explain the various scavenging pumps used in two stroke engine. 4

5. (a) Explain the working of ionisation detector used to measure HC- emission in the exhaust. 8

- (b) Discuss the emissions from CI engines. What are the factors which effect CI engine emissions? 8

6. (a) Draw a neat diagram of thermostatic controlled water cooling system and explain its working. 8

- (b) What is meant by dry and wet sump lubrication? Where dry sump is preferred and why? 8

7. Write short notes 4X4

- (i) Quel fuel and multi fuel engines
- (ii) HUCR dopes
- (iii) Types of super charging
- (iv) Free piston engines.